

DSS Talent Acquisition

NBA Player Evaluation &
Performance Decision Support

GROUP MEMBERS

1. _____
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SPORT ANALYTICS
PLAYER VALUATION

Presentation Agenda

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APA-formatted academic and industry citations

SECTION 01

NBA Positions & Team Roster

Player roles and the 30 NBA franchises



NBA Playing Positions

PG

Point Guard

Primary ball handler and floor general. Leads fast breaks, distributes the ball, sets the offensive tempo, and is the primary free-throw shooter on many teams. Key stats: AST, STL, AST%.

SG

Shooting Guard

Perimeter scoring specialist. Excels at shooting off screens, catch-and-shoot situations, and ball-handling in the backcourt. Often the team's second scoring option. Key stats: PTS, 3P%, eFG%.

SF

Small Forward

Versatile two-way wing player. Can score inside and outside, defend multiple positions, and transition between guard and forward assignments. Key stats: PTS, TRB, BPM.

PF

Power Forward

Interior presence with extended range. Combines rebounding and mid-range/post scoring with the modern ability to stretch the floor via three-point shooting. Key stats: TRB, ORB%, WS.

C

Center

Anchor of the paint on both ends. Primary rebounder, interior scorer, shot blocker, and screen setter. Sets screens to free teammates. Key stats: BLK%, TRB, PER.

NBA Franchises (2024-25)

EASTERN CONFERENCE (15 teams)

ATLANTIC DIVISION

- Boston Celtics
- Brooklyn Nets
- New York Knicks
- Philadelphia 76ers
- Toronto Raptors

CENTRAL DIVISION

- Chicago Bulls
- Cleveland Cavaliers
- Detroit Pistons
- Indiana Pacers
- Milwaukee Bucks

SOUTHEAST DIVISION

- Atlanta Hawks
- Charlotte Hornets
- Miami Heat
- Orlando Magic
- Washington Wizards

WESTERN CONFERENCE (15 teams)

NORTHWEST DIVISION

- Denver Nuggets
- Minnesota Timberwolves
- Oklahoma City Thunder
- Portland Trail Blazers
- Utah Jazz

PACIFIC DIVISION

- Golden State Warriors
- LA Clippers
- Los Angeles Lakers
- Phoenix Suns
- Sacramento Kings

SOUTHWEST DIVISION

- Dallas Mavericks
- Houston Rockets
- Memphis Grizzlies
- New Orleans Pelicans
- San Antonio Spurs

SECTION 02



Data Sources

Three databases powering the DSS analytics engine

Database Overview

DB1

NBA Player Stats (Kaggle)

Records: ~528 rows

Fields: 31 columns

Mid-season snapshot (~30 games per player).
Per-game box-score averages. Joined to BRef
for PNS (KPI 6).

Supports: KPI 3, 6

DB2

NBA Player Salaries 2024-25

Records: ~563 rows

Fields: 3 columns

Full-season salary data. Salary field formatted
as currency string — requires cleaning. Name
normalization (accent stripping) required to
join with BRef.

Supports: KPI 3, 7

DB3

Basketball Reference Advanced Stats

Records: 736 rows

Fields: 30 columns

Full 2024–25 season. Primary source for ALL
advanced metrics. Traded players appear once
per team — deduplicate by keeping highest-G
row per player. Filter '2TM' rows.

Supports: All KPIs

DB1: nba_stats_2024.csv — Variables (Part 1 of 2)

Source: Kaggle · 528 rows · Per-game box-score averages · Mid-season snapshot (≈30 games)

Variable	Description
Rk	Player rank in the dataset (integer identifier, not a KPI metric)
Player	Full player name (used as join key — accent normalization required to match BRef)
Age	Player age at the start of the 2024–25 season
Team	Three-letter team abbreviation (e.g., LAL = Los Angeles Lakers)
Pos	Playing position: PG, SG, SF, PF, or C
G	Games played (used as a filter: $G \geq 30$ or $G \geq 40$ depending on the KPI)
GS	Games started during the season
MP	Minutes played per game (used as a filter: $MP \geq 15$ for qualified-player threshold)
FG	Field goals made per game (counts 2-pointers and 3-pointers)
FGA	Field goal attempts per game
FG%	Field goal percentage = FG / FGA
3P	Three-point field goals made per game
3PA	Three-point attempts per game

DB1: nba_stats_2024.csv — Variables (Part 2 of 2)

Source: Kaggle · 528 rows · Per-game box-score averages · Mid-season snapshot (≈30 games)

Variable	Description
FT	Free throws made per game
FTA	Free throw attempts per game
FT%	Free throw percentage = FT / FTA
ORB	Offensive rebounds per game (used in PNS composite; inputs KPI 6)
DRB	Defensive rebounds per game
TRB	Total rebounds per game = ORB + DRB (used as input to KPI 6 weighted score)
AST	Assists per game (used as input to KPI 6 weighted score)
STL	Steals per game
BLK	Blocks per game
TOV	Turnovers per game (negative input in PER formula from BRef)
PF	Personal fouls per game (negative input in PER formula from BRef)
PTS	Points per game (used as input to KPI 6 Positional Need Score)
Awards	Season award mentions (All-NBA, MVP voting, etc.) — categorical, not used numerically
—	 NOTE: This file does NOT contain advanced metrics (PER, WS, BPM, VORP, USG%, TS%). All advanced analytics are sourced exclusively from Database 3 (Basketball Reference).

DB2: NBA Player Salaries 2024-25 — Variables

Source: Kaggle · ~563 rows · One record = one player's 2024–25 annual salary

Variable	Description
Player	Full player name — join key. Requires accent normalization (unicodedata NFD stripping) to match Basketball Reference names (e.g., 'Jokic' vs. 'Jokić').
Team	Team abbreviation — matches team codes used across all three databases.
Salary	Annual salary in USD, formatted as a currency string (e.g., '\$55,761,216 '). ETL must strip '\$', commas, and trailing spaces then convert

ETL Considerations & Data Quality Notes

- Salary string cleaning: strip leading '\$', remove all commas, strip trailing spaces → cast to integer
- Name join: 334 players match after accent normalization (unicodedata.normalize('NFD') + strip combining marks)
- Max-contract distortion: NBA CBA salary caps cause top players to appear systematically 'undervalued' in KPI 7 — this must be documented as a known model limitation
- No duplicate handling needed: one record per player (unlike BRef which may have multiple rows for traded players)
- KPI use: this file is joined to BRef for Cost-Per-Win (KPI 3) and Salary vs. Market Value Gap (KPI 7) calculations

DB3: Basketball Reference — Variables (Part 1 of 2)

Source: [basketball-reference.com](https://www.basketball-reference.com) · 736 rows · Full 2024–25 season · Primary source for all advanced metrics

Variable	Description
Rk	Row rank (integer; not a performance metric)
Player	Player name — may include Unicode accented characters (e.g., Jokić, Šengün, Dončić); normalize before joining
Age	Player age at start of season (used in KPI 5 age-cohort grouping and KPI 7 regression)
Team	Team code. '2TM' = season total for a player traded mid-season — must be FILTERED OUT before any team-level analysis
Pos	Playing position (PG, SG, SF, PF, C)
G	Games played (filter threshold: $G \geq 30$ or $G \geq 40$ depending on KPI)
GS	Games started
MP	Minutes played per game (filter threshold: $MP \geq 15$ for qualified-player set)
PER	Player Efficiency Rating — Hollinger formula, normalized to 15.0 league average. Primary KPI 1 metric.
TS%	True Shooting % = $PTS / (2 \times (FGA + 0.44 \times FTA))$ — most accurate measure of scoring efficiency; used in KPI 5
3PAr	Three-Point Attempt Rate = $3PA / FGA$ — measures reliance on the 3-point shot
FTr	Free Throw Attempt Rate = FTA / FGA — measures ability to draw fouls
ORB%	Offensive Rebounding % — estimated % of available offensive rebounds a player grabbed while on the floor

DB3: Basketball Reference — Variables (Part 2 of 2)

Source: basketball-reference.com • 736 rows • Full 2024–25 season • Primary source for all advanced metrics

Variable	Description
TOV%	Turnover % — estimated turnovers per 100 plays used (lower is better)
USG%	Usage Rate — estimated % of team plays used by the player while on the floor (KPI 7 regression predictor)
OWS	Offensive Win Shares — wins contributed via offensive production (used as component of WS in KPI 2)
DWS	Defensive Win Shares — wins contributed via defensive production (used as component of WS in KPI 2)
WS	Win Shares = OWS + DWS — cumulative seasonal wins contributed (primary KPI 2 metric; KPI 3 denominator)
WS/48	Win Shares per 48 minutes — normalizes WS for role players vs. high-usage stars; used in KPI 2 and KPI 5
OBPM	Offensive Box Plus/Minus — offensive impact vs. league average per 100 possessions (component of KPI 4)
DBPM	Defensive Box Plus/Minus — defensive impact vs. league average per 100 possessions (component of KPI 4)
BPM	Box Plus/Minus = OBPM + DBPM — all-in-one impact metric; primary KPI 4 metric (Myers, 2015)
VORP	Value Over Replacement Player — contribution above a replacement-level player, scaled to 82 games (KPI 2)
Awards	Season award mentions — categorical label, not used numerically in calculations
—	 ETL: Deduplicate by keeping highest-G row per player. Filter rows where Team = '2TM' (mid-season trade totals) before any team-level grouping (KPI 6). 334–355 players match salary CSV after name normalization.

SECTION 03



KPI Identification

Seven metrics driving evidence-based player acquisition decisions

KPI Overview – Seven Key Metrics

#	KPI Name	Primary Use	Data Source
1	Player Efficiency Rating (PER)	Overall player evaluation filter	BRef
2	Win Shares (WS) / VORP	Cumulative and marginal win contribution	BRef
3	Cost-Per-Win Contribution	Salary efficiency & market OLS regression	BRef + Salary
4	Box Plus/Minus (BPM)	Context-adjusted two-way impact	BRef
5	Performance Trajectory	Age-curve risk and development flags	BRef
6	Positional Need Score (PNS)	Team roster gap analysis by position	BRef + Stats
7	Salary vs. Market Value Gap	OLS residual undervaluation screening	BRef + Salary

KPI 1 — Player Efficiency Rating (PER)

Definition

Developed by Hollinger (2002), PER is a per-minute performance metric that aggregates all positive and negative box-score contributions into a single normalized value. Positive events (FG, FT, 3P, AST, STL, BLK, ORB, DRB) are weighted empirically; negative outcomes (missed shots, TOV, PF) are subtracted. The formula is pace-adjusted for cross-team comparability. PER is normalized so the league average always equals 15.0, providing a stable year-over-year benchmark (Kubatko et al., 2007). Source: Basketball Reference, pre-computed.

Formula / Calculation

$$\text{uPER} = (1/\text{MP}) \times [\text{FG} \times 85.91 + \text{STL} \times 53.90 + 3\text{P} \times 51.76 + \text{FT} \times 46.85 + \text{BLK} \times 39.19 + \text{ORB} \times 39.19 + \text{AST} \times 34.68 + \text{DRB} \times 14.71 - \text{PF} \times 17.17 - (\text{FTA} - \text{FT}) \times 20.09 - (\text{FGA} - \text{FG}) \times 39.19 - \text{TOV} \times 53.90]$$

$$\text{PER} = \text{uPER} \times (15 / \text{lgPER})$$

Filter: $G \geq 30$, $MP \geq 15$

Acquisition threshold: $\text{PER} \geq 18.0$

KPI 1 — Player Efficiency Rating (PER) (cont.)

Acceptable Benchmarks

< 10.0 → Below replacement
Fringe — not an acquisition target

10.0–14.9 → Rotation player
Backup depth; acquisition only on team need

15.0–19.9 → Reliable starter
League average and above; viable target

≥ 20.0 → All-Star / star
Top-10% of qualified players (p90 = 20.0)

≥ 25.0 → MVP-tier
Fewer than 3 players reach this annually

Impact on Business Need

PER serves as the primary screening metric for player evaluation (Business Need 1), enabling scouts to rank large candidate pools by composite efficiency without reviewing individual statistics manually. It anchors the Cost-Per-Win regression in KPI 3 and the prime acquisition filter in KPI 7 ($PER \geq 16 + \text{Gap} < -\$1M$). (Alamar, 2013)

Key Consideration

PER rewards statistical accumulation and does not control for lineup context or pace beyond team-level adjustment. High-usage, high-minute players may inflate PER relative to efficient role players. Always cross-reference with BPM (KPI 4) for context-adjusted validation.

KPI 2 — Win Shares (WS) / VORP

Definition

Win Shares (WS), developed by Oliver (2004) and extended at Basketball Reference, estimates the total team wins a player generated by linking individual efficiency ratings to team performance. WS decomposes into Offensive Win Shares (OWS) and Defensive Win Shares (DWS). WS/48 normalizes cumulative impact per 48 minutes, making it suitable for role-player comparisons. Value Over Replacement Player (VORP) extends this by measuring contribution above a hypothetical replacement-level player (BPM floor of -2.0), scaled to 82 games (Myers, 2011). Together they provide cumulative-season and per-minute signals.

Formula / Calculation

$$WS = OWS + DWS$$

OWS derived from individual offensive rating and team pace.

DWS derived from defensive rating and defensive stops.

$$VORP = [BPM - (-2.0)] \times (\% \text{ team MP}) \times (\text{games} / 82)$$

$$WS/48 = WS / (MP / 48)$$

All pre-computed in BRef.

Filter: $G \geq 40$, $MP \geq 15$

Acquisition: $VORP \geq 1.0$

KPI 2 — Win Shares (WS) / VORP (cont.)

Acceptable Benchmarks

VORP < 0.0 → Below replacement
Player worse than freely available alternatives

WS 2.0–5.0 → Solid rotation
Median player; minimal acquisition premium

WS ≥ 5.0 / VORP ≥ 1.0 → Quality starter
75th percentile in 2024–25 qualified pool

WS ≥ 8.0 / VORP ≥ 4.0 → Elite
Top 5–6% of qualified players; franchise-level

Impact on Business Need

WS and VORP quantify the marginal wins a candidate acquisition adds to the roster, directly addressing team gap analysis (Business Need 3). VORP's replacement-level framing also supports risk assessment (Business Need 4): declining VORP signals diminishing returns. WS/48 supports value identification (Business Need 2) by surfacing efficient role players. (Berri et al., 2006)

Key Consideration

WS is a cumulative metric — players who miss games (injury) accumulate fewer WS without reflecting their per-game quality. Always use WS/48 as the secondary metric for injured players. VORP is sensitive to minutes allocation and team context.

KPI 3 — Cost-Per-Win Contribution

Definition

Cost-Per-Win operationalizes the Moneyball market efficiency principle (Berri et al., 2006): it computes the dollar cost of each Win Share generated. Players producing more wins per salary dollar than peers are acquisition targets (Winston, 2012). A secondary OLS regression model predicts market salary from performance inputs, with the residual identifying over- or undervalued players. This KPI requires joining Basketball Reference (WS, PER, USG%, Age) with the salary CSV, requiring accent-normalized name matching. Filter: $G \geq 40$, $WS > 0$.

Formula / Calculation

$\text{Cost-Per-Win} = \text{Annual Salary} \div \text{Win Shares}$

OLS Regression ($R^2 = 0.550$, $n = 334$):

Predicted Salary = $-\$42.9\text{M}$

+ (PER $\times$ $\$77\text{K}$)

+ (WS $\times$ $\$1.41\text{M}$)

+ (USG% $\times$ $\$1.11\text{M}$)

+ (Age $\times$ $\$1.10\text{M}$)

Salary Gap = Actual – Predicted

Sources: BRef + Salary CSV

(accent normalization required for join)

KPI 3 — Cost-Per-Win Contribution (cont.)

Acceptable Benchmarks

< \$2.0M / WS → Elite value

Top minimum/rookie-scale contracts; priority targets

\$2.0–5.0M / WS → Good value

Solid rotation deals; sustainable cap allocation

\$5.0–10.0M / WS → Pricey

Justifiable only for consistent starters

> \$10.0M / WS → Overvalued

Avoid; poor cap efficiency, limits flexibility

Impact on Business Need

Cost-Per-Win is the central output for value identification (Business Need 2) and salary cap efficiency planning. By ranking players by wins per dollar rather than raw performance, the PDSS enables financially sustainable roster construction under the NBA's hard salary cap (~\$141M) with associated luxury-tax penalties (Coon, 2024).

Key Consideration

NBA max-contract rules create a ceiling on elite salaries, causing top players to systematically appear 'undervalued' in both Cost-Per-Win and OLS gap metrics. This is a known structural limitation — flag elite players separately rather than treating negative gaps as acquisition opportunities.

KPI 4 — Box Plus/Minus (BPM)

Definition

Developed by Myers (2015), BPM estimates a player's contribution to team point differential per 100 possessions relative to a league-average player, using only box-score data. It decomposes into Offensive BPM (OBPM) and Defensive BPM (DBPM); $BPM = OBPM + DBPM$. A BPM of +3.0 means the team outscores opponents by 3 additional points per 100 possessions with that player on the court. Unlike PER, BPM is calibrated against Regularized Adjusted Plus/Minus (RAPM) data to control for lineup context (Engelmann, 2017). Replaces FIFA Overall Rating as the DSS composite impact metric.

Formula / Calculation

$BPM = f(\text{role, per-possession box stats, team context})$
Calibrated against RAPM as ground truth.

$$BPM = OBPM + DBPM$$

OBPM: scoring + playmaking impact
DBPM: defensive value

All three columns pre-computed in BRef.

Filter: $G \geq 30$, $MP \geq 15$

Acquisition threshold: $BPM \geq +2.0$

Two-way elite: $OBPM > +1.5$ AND $DBPM > +1.5$

KPI 4 — Box Plus/Minus (BPM) (cont.)

Acceptable Benchmarks

< -2.0 → Below replacement

Not an acquisition candidate in any scenario

-1.0 to +1.0 → Average rotation

Serviceable depth; situational value only

+1.0 to +4.0 → Quality starter

Confirmed productive contributor; acquisition target

≥ +4.0 → All-Star / Elite

≈ Top 7% of qualified players; highest-priority target

Impact on Business Need

BPM complements PER (Business Need 1) by providing a context-adjusted efficiency view accounting for possession-level game situations. The DBPM component identifies undervalued defensive specialists — a confirmed NBA market inefficiency (Ilardi & Barzilai, 2008). Combined with VORP, BPM ensures the shortlist captures full winning contributions beyond scoring.

Key Consideration

BPM is more conservative than PER and may undervalue elite scorers in certain team systems. DBPM specifically can be unstable for players on teams with weak defensive schemes. Always review both OBPM and DBPM separately — a high BPM masking a very low DBPM signals an offensive-only player.

KPI 5 — Performance Trajectory

Definition

Performance Trajectory measures whether a player is performing above, at, or below the expected output for their age cohort across PER, WS/48, and TS%. Since only single-season data is available, the DSS implements an age-curve proxy derived from basketball aging-curve research (Albert et al., 2017; Bradbury, 2009): each player's metrics are compared against league-average values for their age group (≤ 22 , 23–25, 26–28, 29–31, 32+), with deviation from the cohort norm used as the trajectory signal. The output is directional, not predictive.

Formula / Calculation

Age Cohorts (from BRef 2024–25):
 ≤ 22 / 23–25 / 26–28 / 29–31 / 32+

For each cohort: compute league avg PER

$\Delta \text{PER} = \text{Player PER} - \text{Cohort Mean PER}$

Rising flag: Age ≤ 26 AND $\Delta \text{PER} \geq +1.5$

Peak: Ages 26–29 (reference window)

Decline flag: Age ≥ 30 AND $\Delta \text{PER} \leq -1.5$

Risk flag: Age ≥ 32 AND declining

Secondary: TS% deviation $\geq +0.10$ above cohort norm

KPI 5 — Performance Trajectory (cont.)

Acceptable Benchmarks

▲ Rising — Age ≤ 26 , Δ PER $\geq +1.5$

~20% of qualified pool; prioritize multi-year deals

● Stable — Δ PER within ± 1.0

~60% of pool; reliable but limited upside

▼ Declining — Age ≥ 30 , Δ PER ≤ -1.5

~10% of pool; avoid multi-year contract exposure

⚠ Risk — Age ≥ 32 AND declining

Survivorship bias inflates cohort avg; use caution

Impact on Business Need

Performance Trajectory is the primary indicator for risk assessment (Business Need 4), flagging players whose current statistics overstate expected future production. It also refines value identification (Business Need 2) by distinguishing genuinely undervalued players from those whose modest salaries reflect a market-recognized downward trajectory. Critical for multi-year deal evaluation.

Key Consideration

The 32+ cohort average PER is inflated by survivorship bias — only elite players remain active past 32. This means NEGATIVE deviations in that cohort are more significant than the formula suggests. Multi-season BRef data (2022–23 through 2024–25) would enable true YoY trajectory calculation and should be incorporated as the system matures.

KPI 6 — Positional Need Score (PNS)

Definition

PNS is a derived composite metric quantifying the statistical deficiency of a team's positional group relative to the league average. Constructed using a weighted z-score approach grounded in Multi-Criteria Decision Analysis (Saaty, 1980), PNS computes the gap between a team's position-group averages and league-wide norms for four performance dimensions: PER (40%), PTS/g (25%), REB/g (20%), and AST/g (15%). Normalized to a 0–3 scale for cross-team interpretability. Requires joining BRef (PER) with Kaggle Stats CSV (PTS, TRB, AST).

Formula / Calculation

For each Team T and Position P:

$$z_metric(T,P) = \frac{[League\ Avg\ at\ P - Team\ Avg\ at\ P]}{\div\ League\ SD\ at\ P}$$

$$PNS(T,P) = 0.40 \times z_PER \\ + 0.25 \times z_PTS \\ + 0.20 \times z_REB \\ + 0.15 \times z_AST$$

$$Normalized: PNS_norm = (PNS - min) / (max - min) \times 3$$

League baselines by Pos (PER | PTS | REB | AST):

PG: 14.8 | 13.2 | 3.4 | 5.0 SG: 13.1 | 11.2 | 3.1 | 2.4

SF: 13.4 | 11.5 | 4.0 | 2.2 PF: 14.9 | 11.4 | 5.2 | 2.1

C: 18.4 | 11.2 | 7.3 | 2.1

KPI 6 — Positional Need Score (PNS) (cont.)

Acceptable Benchmarks

< 0.5 → Adequately staffed

No immediate acquisition action needed at this position

0.5–1.5 → Minor gap

Monitor; prioritize in next offseason if persists

1.5–2.5 → Moderate need

47.7% of all team-position combinations in 2024–25

≥ 2.5 → Critical need

3.9% of combinations; immediate acquisition priority

Impact on Business Need

PNS directly addresses team gap analysis (Business Need 3), ensuring acquisition decisions are driven by roster composition logic rather than absolute player rankings alone. Without PNS, a DSS might recommend an excellent player at an already strong position while missing a critical roster weakness. Combined with PER and VORP filters, PNS ensures the shortlist maximizes win impact at positions of greatest need. (Alamar, 2013)

Key Consideration

Center PNS baselines are inflated (avg PER 18.4) due to survivorship bias — only elite centers play significant minutes. Apply position-specific league averages as the reference, not a universal threshold. ETL must filter '2TM' rows from BRef and match accent-normalized player names before computing team-level groupings.

KPI 7 — Salary vs. Market Value Gap

Definition

Market value is estimated using an OLS regression model predicting fair-market annual salary from four performance inputs: PER, Win Shares, Usage Rate, and Age. The Salary Gap (actual minus predicted) identifies undervalued players (negative gap = acquisition targets) and overvalued ones (positive gap = avoid). This operationalizes the labor market efficiency framework of Berri & Krautmann (2006) and extends Winston's (2012) quantitative salary modeling to the current NBA market. Requires joining BRef with salary CSV (accent normalization; 355 matched players).

Formula / Calculation

Step 1 — OLS Regression ($R^2 = 0.550$, $n = 355$):

Predicted Salary = $-\$42.9\text{M}$

+ (PER $\times$ $\$77\text{K}$)

+ (WS $\times$ $\$1.41\text{M}$)

+ (USG% $\times$ $\$1.11\text{M}$)

+ (Age $\times$ $\$1.10\text{M}$)

Step 2 — Salary Gap:

Gap = Actual Salary – Predicted Salary

Negative Gap $\rightarrow$ Undervalued target

Positive Gap $\rightarrow$ Overvalued — avoid

Residual SD $\approx$ $\$8.9\text{M}$

Filter: $G \geq 40$, $MP \geq 15$

Prime filter: Gap $< -\$1\text{M}$ AND PER ≥ 16

KPI 7 — Salary vs. Market Value Gap (cont.)

Acceptable Benchmarks

< -\$3.0M → High-priority undervalued

~39% of matched pool; prime acquisition targets

-\$1M to -\$3M → Moderate undervaluation

Directional signal; validate with PER ≥ 16 filter

±\$1.0M → Fair market value

~12% of pool; model predicts salary accurately

> +\$3.0M → Overpriced

~30% of pool; flag especially if Trajectory declining

Impact on Business Need

The Salary Gap is the most direct operationalization of value identification (Business Need 2). Combined with PER ≥ 16, it produces the most actionable acquisition shortlist in the DSS — financially disciplined candidates balancing performance impact against cap sustainability. Players with large positive gaps AND KPI 5 decline flags represent the most dangerous contracts to inherit in a trade. (Berri & Krautmann, 2006)

Key Consideration

NBA max-contract rules create a structural ceiling: elite players (e.g., those on max deals) will always appear 'undervalued' in this model regardless of actual market standing. Document this limitation explicitly. The $R^2 = 0.550$ means 45% of salary variance lies outside box-score data (market size, name recognition, contract timing).

SECTION 04

Requirements & Tasks

Six analytical workflows powering the acquisition DSS



Requirements / Tasks – Overview

#	Requirement / Task	KPIs Supported
R1	Player Performance Filtering	PER, WS, BPM
R2	Salary-Value Alignment Analysis	Cost-Per-Win, Salary Gap
R3	Statistical Analogue Matching	PER, BPM, WS/48, Trajectory
R4	Performance Trajectory & Risk	Trajectory, VORP, WS/48
R5	Positional Gap Analysis	PNS
R6	Integrated Acquisition Recommendation	All 7 KPIs

Requirements Detail – R1, R2 & R3

R1

Player Performance Filtering

KPIs: PER | WS | BPM

Input: BRef (PER, WS, BPM, WS/48, G, MP, Pos, Age) — deduplicated; keep highest-G row per player

Action: Filter: $G \geq 30$ and $MPG \geq 15$. Rank by weighted composite: $PER(40\%) + BPM(30\%) + WS/48(30\%)$. Apply $PER \geq 18$ threshold.

Output: Ranked shortlist of up to 94 candidates with composite score, position, age-group flag, and trajectory indicator

R2

Salary-Value Alignment Analysis

KPIs: Cost-Per-Win | Salary Gap

Input: BRef (WS, PER, USG%, Age) joined with Salary CSV. Filter: $G \geq 40$, $WS > 0$

Action: Compute $\text{Cost-Per-Win} = \text{Salary} / \text{WS}$. Run OLS regression to predict market salary. Calculate $\text{Gap} = \text{Actual} - \text{Predicted}$. Apply tiers: $< \$2M$ =green, $\$2M$ – $5M$ =yellow, $> \$10M$ =red.

Output: Color-coded value matrix with tier, Cost-Per-Win, actual vs. predicted salary, and gap; prime targets flagged ($\text{Gap} < -\$1M$ AND $PER \geq 16$)

R3

Statistical Analogue Matching

KPIs: PER | BPM | WS/48 | Trajectory

Input: BRef (PER, WS/48, BPM, USG%, TS%, Age) + Salary CSV for cost comparison

Action: Apply k-means clustering on normalized PER, WS/48, BPM, USG%, TS% vectors. For a target player profile, identify 5–10 nearest clusters at lower salary. Rank by Cost-Per-Win differential.

Output: List of cost-effective statistical comparables with side-by-side KPI comparison, salary, gap, and age-trajectory flag per analogue

Requirements Detail — R4, R5 & R6

R4

Performance Trajectory & Risk Assessment

KPIs: Trajectory | VORP | WS/48

Input: BRef (PER, WS/48, TS%, Age, G, MP) — age-curve norms computed from same dataset by age bucket (≤ 22 , 23–25, 26–28, 29–31, 32+)

Action: Compute PER deviation vs. age-group norm. Flag: green if Age ≤ 26 and $\Delta \geq +1.5$; red if Age ≥ 30 and $\Delta \leq -1.5$; orange if Age ≥ 30 and $\Delta < 0$. Apply survivorship-bias note for 32+ cohort.

Output: Risk-tiered player list (green/orange/red) with PER deviation, age bucket, WS/48 trend, and salary from salary CSV for financial exposure context

R5

Positional Gap Analysis

KPIs: Positional Need Score (PNS)

Input: nba_stats_2024.csv (PTS, TRB, AST, Team, Pos) joined with BRef (PER). Filter: MPG ≥ 15 , G ≥ 20 . Exclude '2TM' rows from BRef.

Action: Group by Team and Pos. Compute weighted z-score gap vs. league norms for PER(40%), PTS(25%), REB(20%), AST(15%). Normalize to 0–3 scale. Positions with PNS ≥ 1.5 flagged as acquisition priorities.

Output: Positional heatmap (PG/SG/SF/PF/C $\times$ all 30 teams) color-coded by PNS severity; critical (≥ 2.5) and moderate (1.5–2.4) positions highlighted for the target team

R6

Integrated Acquisition Recommendation

KPIs: All 7 KPIs

Input: Outputs from R1–R5: performance shortlist, value matrix, risk tiers, PNS heatmap, and analogue list — cross-referenced by player name

Action: Normalize each KPI output to 0–1 scale. Apply user-configurable weights (default: PER 20%, Gap 20%, BPM 15%, VORP 15%, PNS fit 15%, Trajectory 15%). Compute final composite Acquisition Score.

Output: Final ranked acquisition shortlist: composite score, Cost-Per-Win, salary gap, risk tier, positional fit indicator, and trajectory flag per candidate; top 10 highlighted for immediate action

REFERENCES



Alamar, B. C. (2013). Sports analytics: A guide for coaches, managers, and other decision makers. Columbia University Press.

Alamar, B. C., & Mehrotra, V. (2011). Beyond Moneyball: The rapidly evolving world of sports analytics, part 1. Analytics Magazine, 4(5), 4–6.

Albert, J., Glickman, M. E., Swartz, T. B., & Koning, R. H. (Eds.). (2017). Handbook of statistical methods and analyses in sports. CRC Press.

Berri, D. J., & Krautmann, A. C. (2006). Shirking on the court. Economic Inquiry, 44(3), 536–546. <https://doi.org/10.1093/ei/cbj033>

Berri, D. J., Schmidt, M. B., & Brook, S. L. (2006). The wages of wins. Stanford University Press.

Bradbury, J. C. (2009). Peak athletic performance and ageing. Journal of Sports Sciences, 27(6), 599–610.

Cervone, D. et al. (2016). A multiresolution stochastic process model. JASA, 111(514), 585–599.

Coon, L. (2024). NBA salary cap FAQ. cbafaq.com

Engelmann, J. (2017). Possession-based player performance analysis. MIT Sloan Sports Analytics Conference.

Hollinger, J. (2002). Pro basketball forecast, 2002–03. Brassey's.

Ilardi, S., & Barzilai, A. (2008). Adjusted plus/minus ratings. 82games.com

Kubatko, J. et al. (2007). A starting point for analyzing basketball statistics. JQAS, 3(3). <https://doi.org/10.2202/1559-0410.1070>

Myers, D. (2011). VORP: A valuable player statistic. basketball-reference.com

Myers, D. (2015). Box plus/minus (BPM). basketball-reference.com

Oliver, D. (2004). Basketball on paper. Potomac Books.

Saaty, T. L. (1980). The analytic hierarchy process. McGraw-Hill.

Winston, W. L. (2012). Mathletics. Princeton University Press.